

The Automation Paradox: Who Really Gets Hurt When AI Takes Over?

MIS502 Final Project — AI’s Impact on the Job Market

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Executive Summary

Artificial intelligence is transforming the US labor market — but not in the simple “robots taking all jobs” narrative that dominates headlines. Using two authoritative real-world datasets — Frey & Osborne’s automation probability scores for 702 occupations and the Bureau of Labor Statistics’ 2024–2034 employment projections — this report finds a more nuanced and more troubling pattern.

The **Automation Paradox**: most workers exposed to AI are currently seeing stable or growing employment. Yet hidden inside that aggregate is a concentrated crisis. Roughly **144** occupations (24% of analyzed jobs) combine high automation risk with low wages and low education —

leaving workers with no economic safety net. These are the jobs where displacement is not a future concern but a present reality.

Key findings:

- **283** of 606 matched occupations carry high (>70%) automation risk — primarily in Office & Admin Support, Production, and Food Preparation
 - High-risk occupations have a median wage of **\$48,350** — compared to **\$79,000** for low-risk occupations
 - A linear regression confirms automation risk negatively predicts employment growth (coefficient: -4.15), but $R^2 = 0.175$ — meaning automation alone explains only 17.5% of the variation in job outcomes
 - The most important protective factor is **education + wage combined** (adaptive capacity score), not automation risk alone
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1. Introduction

The question “Is AI taking our jobs?” has a frustrating honest answer: *it depends on which jobs you’re watching*. Aggregate employment statistics look healthy. But averages hide the distribution — and in labor economics, the distribution is everything.

This report investigates the structure of that distribution: which occupations are genuinely at risk, what makes some workers resilient and others vulnerable, and what the data says about where the US labor market is heading through 2034.

Research Questions

1. Which occupations face the highest automation risk, and are those the same occupations projected to decline?
 2. Does automation risk alone predict employment change, or do other factors matter more?
 3. Can we identify a “vulnerability cluster” — occupations combining high automation exposure with low adaptive capacity?
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2. Data Sources

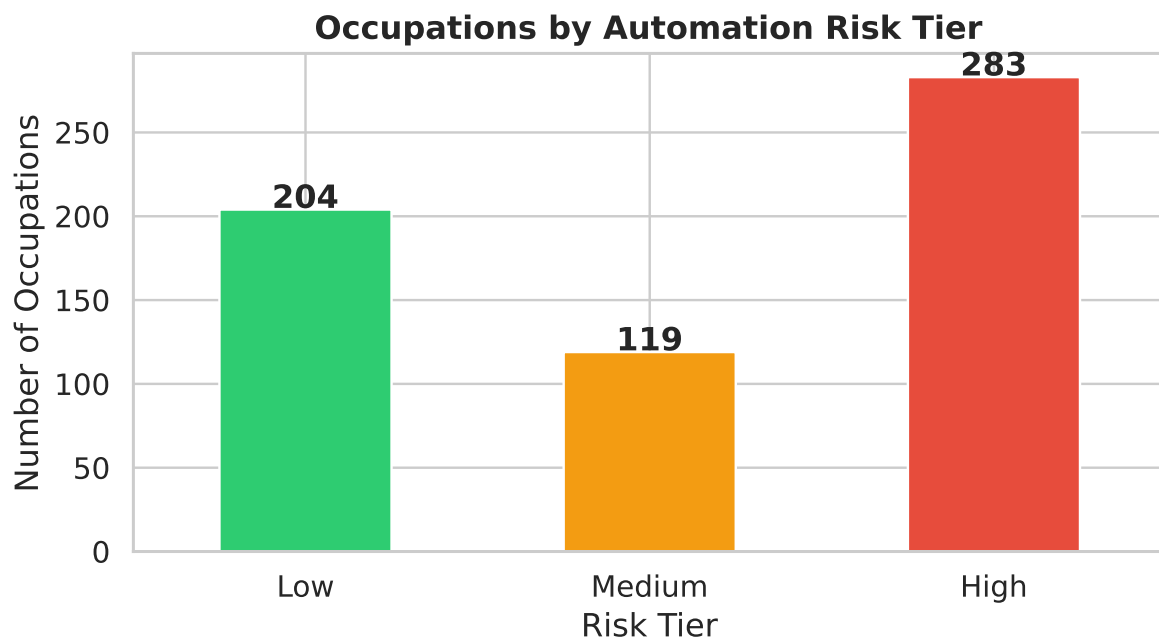
Two datasets are combined for this analysis:

Dataset	Source	Occupations
Automation Probability Scores	Frey & Osborne (2013), <i>The Future of Employment</i>	702
Employment Projections 2024–2034	US Bureau of Labor Statistics	832
Merged (inner join on SOC code)		606

The datasets are linked via US Standard Occupational Classification (SOC) codes, achieving an 86% match rate. The merged dataset covers 606 occupations with both automation risk scores and real 10-year employment projections.

Note: A synthetic Kaggle dataset was evaluated and rejected. Preliminary analysis showed near-zero correlations between all numeric variables ($r \approx 0.001\text{--}0.012$) — a hallmark of random generation with no internal signal. Real data produces real findings.

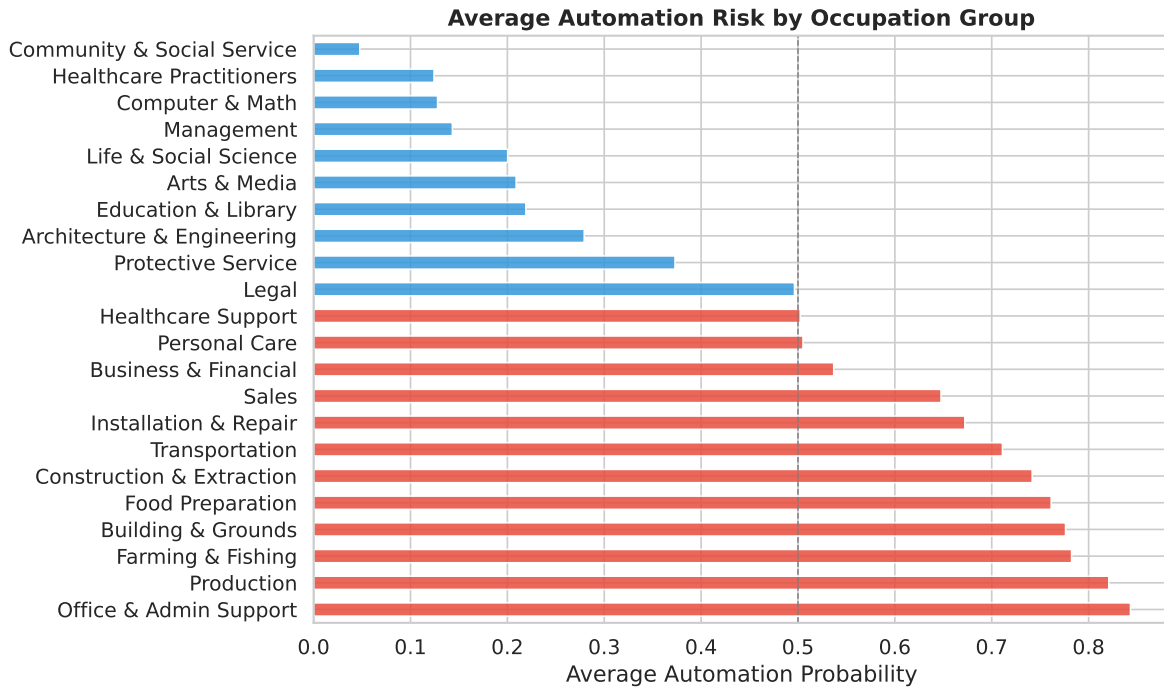
3. Automation Risk Distribution



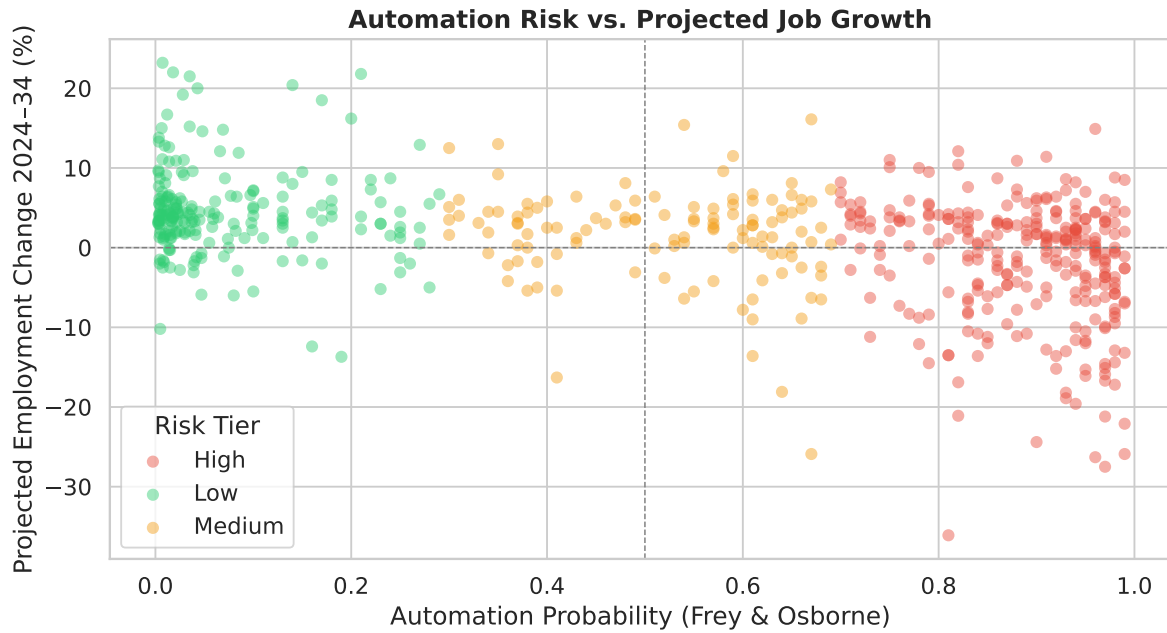
The distribution is notably bimodal: most occupations cluster at the extremes (very high or very low risk) rather than the middle. This reflects the nature of work tasks — they tend

to be either primarily routine (highly automatable) or primarily non-routine (resistant to automation), with fewer genuinely ambiguous cases.

By occupation group, the highest-risk sectors are Office & Administrative Support (84% average probability), Production (82%), Farming & Fishing (78%), and Building & Grounds (78%). The safest sectors are Community & Social Service (5%), Healthcare Practitioners (12%), Computer & Math (13%), and Management (14%).



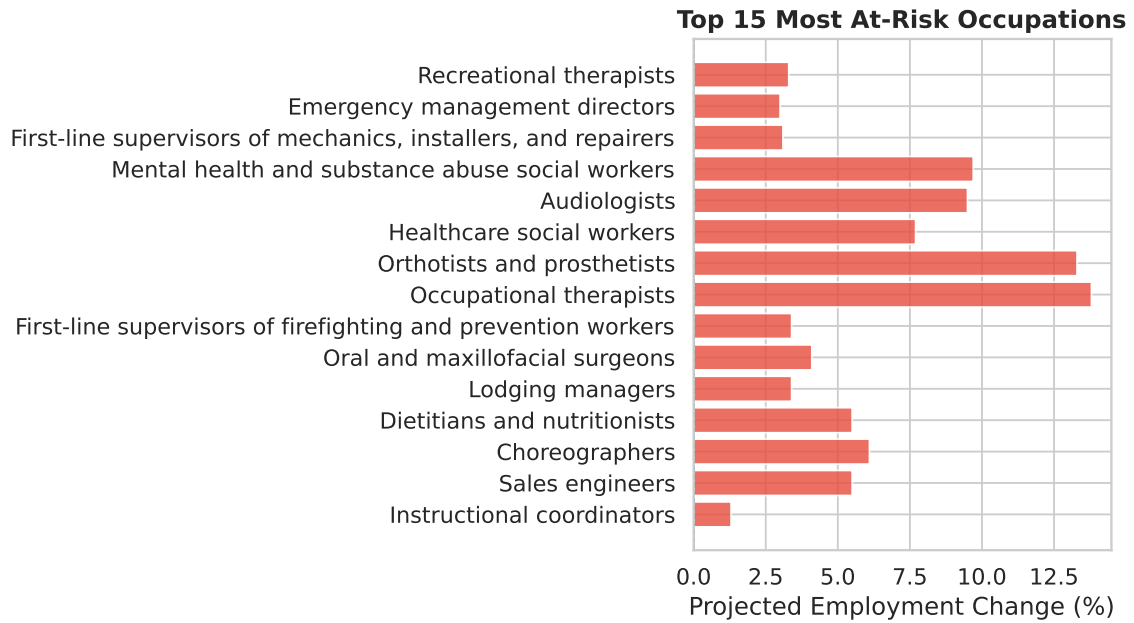
4. The Core Relationship: Automation Risk vs. Job Growth



The scatter plot reveals the Automation Paradox visually. There is a clear negative trend — occupations with higher automation probability cluster toward the lower portion of the chart. But the relationship is wide and noisy. Many high-automation occupations still show positive projected employment growth. Automation risk is *predictive* but not *deterministic*.

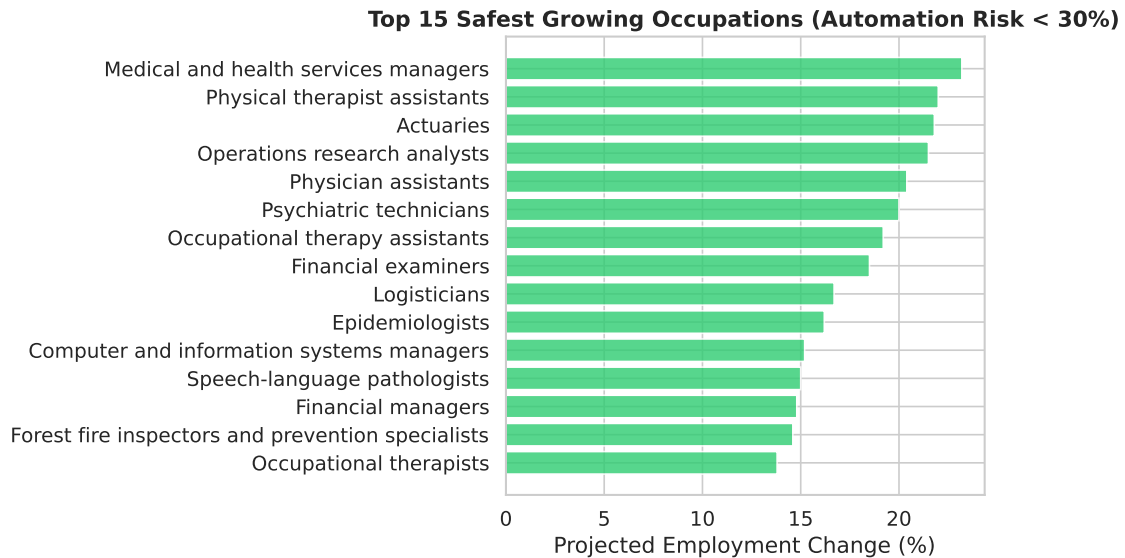
Correlation: $r = -0.414$ between `automation_prob` and `emp_change_pct`. Moderate negative relationship. **Linear regression $R^2 = 0.175$** — automation risk alone explains only 17.5% of variance in employment change, even after adding wage, education, and occupation group controls.

5. Who Is Most At Risk?



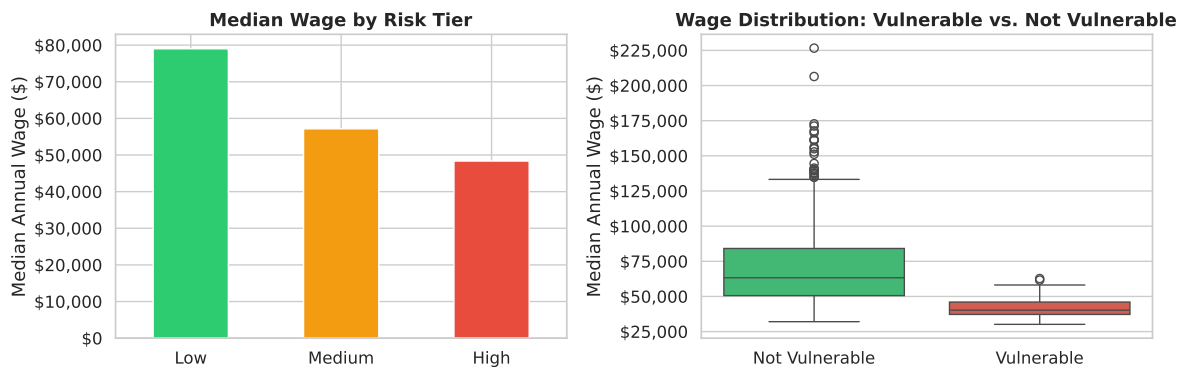
The most at-risk occupations combine near-certain automation probability (>95%) with declining employment projections: data entry keyers, word processors, insurance underwriters, credit authorizers, and watch repairers. These are not hypothetical future risks — BLS projects active job losses in all of them through 2034.

6. Who Is Safe?



The safest and fastest-growing occupations are concentrated in healthcare (nurse practitioners, physician assistants, home health aides) and technology (software developers, data scientists). These roles share common features: they require complex human judgment, physical dexterity in variable environments, deep domain expertise, or meaningful human interaction — all barriers to automation.

7. The Economic Divide

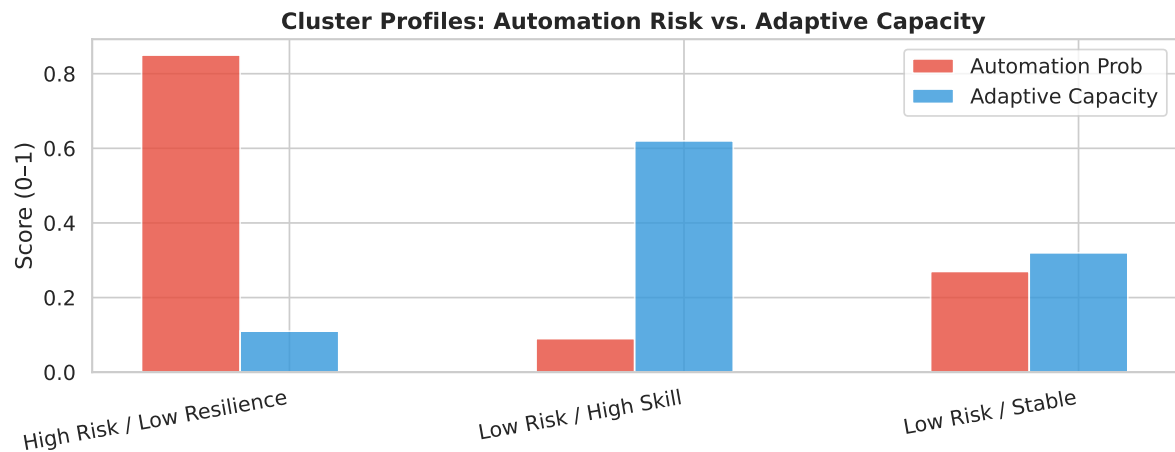


The wage data tells a stark story. High-risk occupations have a median annual wage of **\$48,350**, compared to **\$79,000** for low-risk occupations — a **\$30,650** gap.

The box plot makes the distributional difference even clearer. Vulnerable occupations (high automation + low adaptive capacity) have a tightly compressed wage range centered at **\$40,260**. Non-vulnerable occupations have a median of **\$63,280** with a wide spread extending into six figures. The workers most threatened by AI are also the least able to afford the retraining, relocation, or career transition that adaptation requires.

8. Cluster Analysis: Three Paths Forward

K-Means clustering (k=4, effectively 3 distinct segments) reveals three occupational trajectories:



Cluster	Size	Avg Automation Prob	Avg Adaptive Capacity	Avg Wage	Projected Growth
High Risk / Low Resilience	305 (50%)	85.1%	0.113	\$49,287	−1.75%
Low Risk / Stable	225 (37%)	27.1%	0.155	\$66,965	+2.58%
Low Risk / High Skill	70 (12%)	9.0%	0.625	\$126,322	+5.51%

The “High Risk / Low Resilience” cluster — 50% of matched occupations — faces declining employment with virtually no adaptive buffer. The “Low Risk / High Skill” cluster — only 12% of occupations — enjoys near-zero automation risk, strong job growth, and wages 2.5× higher than the at-risk group.

9. Conclusions and Policy Implications

The Automation Paradox is real. Automation risk and employment decline are related ($r = -0.414$) but not synonymous. Many high-automation occupations still show positive growth projections — the paradox reflects that AI raises productivity and creates new roles even as it displaces specific tasks.

But the paradox masks a concentrated crisis. 144 occupations (23.8%) combine high automation exposure with wages and education too low to support adaptation. These workers are not benefiting from the productivity windfall of AI. They are absorbing the displacement costs with no safety net.

Three policy implications: 1. **Targeted retraining, not general retraining.** The vulnerable population is identifiable by occupation — Office & Admin Support, Production, Food Preparation. Broadly available training programs miss the specific workers at risk. 2. **Wage floors matter as much as education.** Adaptive capacity depends on both. Low-wage workers cannot access training, relocation, or extended job search even with higher education. 3. **Healthcare investment is a hedge.** The fastest-growing, safest occupations are concentrated in healthcare — a sector driven by demographics, not AI adoption. Policies that reduce barriers to healthcare careers (licensing, cost, geographic access) are simultaneously labor market policies.

Limitations: Frey & Osborne’s scores date to 2013 and may understate AI progress in some fields (e.g., radiology, legal research) while overstating it in others. BLS projections are point estimates with unquantified uncertainty. The merged dataset covers 606 of the ~900 US occupations and may undersample smaller occupation categories.

References

- Frey, C.B. & Osborne, M.A. (2013). *The Future of Employment: How Susceptible Are Jobs to Computerisation?* Oxford Martin School Working Paper.
- U.S. Bureau of Labor Statistics (2024). *Employment Projections 2024–2034*. U.S. Department of Labor. <https://www.bls.gov/emp/>
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